

PHARMACOEPIDEMIOLOGY, CHAPTER 42

Patient Engagement and Patient-Reported Outcomes

A Plain-Language Guide

Why measuring health from the patient's own point of view is reshaping clinical care, drug development, and the research used to evaluate both.

Adapted from Chapter 42 of Pharmacoepidemiology, 6th Edition, edited by Brian L. Strom, Stephen E. Kimmel, and Sean Hennessy (Esi M. Morgan, author)
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The Patient's Changing Role in Healthcare

A historical view of the relationship between patient and clinician — physician, provider, or otherwise — positioned the patient as the recipient of a “healthcare product”: clinician-generated information, advice, a treatment prescription, and instructions the patient was expected to follow.

Under this transactional view, the outcome of care depended on just two things: the quality of the clinician's advice, and the patient's ability and inclination to carry it out. This chapter traces a more collaborative model that is reshaping how healthcare is delivered — and, with it, how a patient's own outcomes get measured.

The IOM's Case for Patient-Centered Care

In 2001, an influential report by the Institute of Medicine on reforming and improving US healthcare named patient-centeredness as one of six guiding aims for the system, which should also be safe, effective, timely, efficient, and equitable.

A related, more far-reaching argument goes further: it holds that healthcare is better conceived as a service than a product provided to patients, and that achieving the best outcomes may hinge on taking a co-production approach to care — one in which the healthcare service, and the responsibility for its outcomes, are created jointly by patient and clinician.

From Product to Service: Co-Production

In this service-oriented view, the patient (the service user) and the clinician (the service professional) are in a collaborative relationship, with shared creation of the healthcare service and shared responsibility for its outcomes. Effectively, the healthcare service is “co-produced.”

Self-management support is a key component of effective chronic illness care that is patient centered. Concepts such as effective communication, shared decision making, and monitoring of health outcomes become important, essential goals for co-producing patient-centered care, particularly for chronic conditions.

Patients as Partners in Research

Reframing the patient-clinician relationship as co-producers increases the importance of measuring patient-reported outcomes in healthcare delivery — and, by extension, in the observational studies and clinical trials that evaluate treatments. Patient engagement in research more broadly has become a phenomenon over the past few decades, driven by patient advocacy efforts.

Patients are moving from the role of study subject to becoming partners in prioritizing research questions and contributing to study design. The Patient-Centered Outcomes Research Institute (PCORI) has been funding research since 2012 that promotes the engagement of multiple stakeholders — patients, clinicians, payers, and healthcare institutions — in the conduct of clinical research, and is systematically studying the impact of patient engagement on the research process itself.

What Patient Stakeholders Bring to a Study

Patient stakeholders on a research team can shape a study in ways a purely clinical or academic team might miss.

Shaping the Research Question

Influence which topics get studied, steering priorities toward what matters most to patients and to clinical relevance.

Shaping the Study Itself

Contribute to study design, including which outcome measures are selected, and help design materials such as consent forms.

Strengthening Data Collection

Participate in recruitment and retention efforts, and help reduce missing data over the course of the study.

Interpreting the Results

Provide perspective that helps researchers interpret what a study's findings actually mean for patients.

These process contributions are promising, but at the time of the chapter it was still too early to know whether patient engagement actually improves study completion or the uptake of findings into clinical practice.

What Is a Patient-Reported Outcome?

Patient-reported outcomes, or PROs, are defined as any report of the status of a patient's health condition that comes directly from the patient, without interpretation of the patient's response by a clinician or anyone else.

PROs let patients directly communicate information about their health — symptoms, function, quality of life, participation, health behaviors, and experience with care — in a structured format that translates lived experience into data. They exist because a patient's health concerns may not be adequately perceived by a clinician unless there is direct, structured inquiry into what is actually bothering the patient.

What a PRO Can Capture

Use of PROs recognizes that a clinician cannot fully perceive a patient's health concerns without asking directly and systematically. Failing to elicit that input means clinicians may end up prioritizing health topics that matter less to the patient than to the clinician.

Incorporating PROs into clinical care has the potential to enrich clinician-patient communication: the patient's voice is systematically and routinely collected, and provided as data in the form of meaningful scores. Clinical decisions and interventions can then be anchored to those scores, complementary to lab tests or a physical exam.

Four Ways to Measure a Clinical Outcome

Beyond PROs, clinical research recognizes three other forms of clinical outcome assessment, each defined by who is doing the reporting.

Patient-Reported

Comes directly from the patient, with no interpretation added by a clinician or anyone else — ratings, symptom reports, or questionnaires.

Clinician-Reported

Completed by a healthcare professional based on their evaluation — physical findings or clinical events supported by lab or physiologic tests.

Observer-Reported

Based on observable signs or behaviors, but does not require interpretation by a professional — can be reported by a caregiver, such as a parent.

Performance-Based

Based on the patient completing a task as directed by a professional, such as a timed walk or a memory test.

Building Sharper Measurement Tools

Advances in measurement theory, particularly the use of item response theory (IRT) in scale development, have enabled a new generation of PRO measures that are both more efficient — shorter — and more precise than questionnaires built on older, classical test theory.

In support of clinical research, the National Institutes of Health funded the development of several IRT-based PRO systems, including the Patient-Reported Outcomes Measurement Information System (PROMIS), which started as a cooperative network in 2004. PROMIS measures were designed to be generic and usable across chronic conditions, available as short forms, item banks, and computer adaptive tests, and their use has since evolved from supporting research to supporting outcomes assessment in everyday clinical care.

A Compelling Vision, a Practical Challenge

Treating PROs as an essential component of patient engagement has the power not only to influence and improve outcomes, but potentially to transform how patients perceive their own health and agency, and how they communicate with their healthcare professionals.

The vision is compelling, but the reality of implementing, interpreting, using, and evaluating PROs poses real practical and methodologic challenges. The rest of this chapter explores those challenges in the context of integrating PROs into pharmacoepidemiology research, including both clinical trials and observational studies.

Capturing the Patient's Voice in Drug Trials

It is increasingly recognized that including the perspective of the patient end-user is important during the development of a new medication or medical product.

Capturing the patient perspective as a clinical trial outcome may take the form of a PRO: a rating scale, a symptom report, or a questionnaire completed by self-report or through a structured or unstructured interview — provided there is no modification of the patient's response by the interviewer. PROs can capture symptoms and other domains of health that are otherwise unobservable, and can also capture the patient's own viewpoint on observable domains, such as functional ability or counts of events.

What PROs Reveal About a New Medication

PROs serve a variety of functions in new product development. They can help researchers better understand the impact of a medical condition on a patient's physical, mental, or social health, and can uncover unmet medical needs that a new medication needs to address.

PROs can also be used to estimate the benefit of a medication to the patient during a clinical trial — for example, by estimating the minimal clinically important treatment effect or difference — and, by extension, to characterize the benefit patients can expect once the medication reaches everyday clinical use. Adverse effects of treatment, or an estimate of the risks of use, may also be captured from the patient's own perspective.

The FDA Wants to Hear From Patients Too

As part of the 2012 reauthorization of the Prescription Drug User Fee Act (PDUFA V), the Food and Drug Administration has been including patient input on the impact of chronic illness, to inform which aspects of a disease would be most meaningful to target from the patient's own perspective.

Gathering this kind of information from cohorts of patients with chronic conditions helps identify the disease impacts and treatment benefits that patients themselves consider most meaningful. Using that information well to inform drug development requires a systematic approach and robust interpretive methods.

Statistically Significant Isn't Always Clinically Meaningful

Once a clinical outcome that matters to patients has been identified as a meaningful drug target, the next question is whether a treatment produces a clinically meaningful effect on it — a different question from whether the effect is statistically significant.

A statistically significant change in an outcome measure's score may be achieved without corresponding to a clinically meaningful benefit. Determining that a change is clinically meaningful rests on the idea that the outcome being measured matters to the patient to begin with, and that the degree of change creates a benefit the patient can actually appreciate — a reduction in symptoms, an increase in function, or greater wellbeing. Understanding how to estimate this kind of meaningful change remains an area of active methodological research.

The Trouble With PROs as Real-World Outcomes

A current challenge to using PROs as outcomes in observational studies is that such measures are not typically available, especially compared to measuring other outcomes such as clinical events or death.

PRO measures are nonetheless growing in use in clinical care, in part due to the push for patient engagement to provide patient-centered care and improve healthcare value. Conceptually, capturing PROs in clinical care can be seen as a method for including the patient's voice in how outcomes are estimated — and in practice, incorporating PROs is starting to produce meaningful healthcare improvement.

From Feasibility to Real Impact

Published reports on integrating PROs into clinical care have evolved over time: from early descriptions of acceptability and feasibility, to demonstrations of improved care processes and communication, to preliminary evidence of increased clinician satisfaction and even improved outcomes.

When PROs are collected and reviewed, patients perceive their clinician to have increased awareness of symptoms that might otherwise go unrecognized. Clinicians may be better able to identify conditions, including mental health concerns, and communication, shared decision making, and collaborative treatment planning are all enhanced when PROs are incorporated into patient-clinician conversations.

ROUTINE CARE, 3 OF 4

A Landmark Finding: PROs and Cancer Survival

Much of the evidence on the effective use of PROs in clinical medicine has come from oncology and surgery. A compelling report from an oncology randomized trial of PRO use during routine cancer treatment found that integrating electronic symptom monitoring into the care of patients with metastatic cancer was associated with increased survival.

The authors suggest this may be a consequence of a more responsive care team: earlier detection of symptoms allowed adjustments to chemotherapy dose that averted adverse events, and also allowed patients to stay on effective chemotherapy for a relatively longer duration because of reduced adverse symptoms. Even setting survival aside, the reduction in patient suffering that PROs helped achieve here is meaningful in its own right.

Payers Are Paying Attention

Recently, payers have been encouraging PRO collection, including through the use of financial incentives for tracking PRO data.

For example, PROs are used to evaluate the impact and trajectory of improvement following surgical interventions in the case of Medicare reimbursement for elective joint replacement. The availability of this kind of data can enhance observational studies by providing prospectively collected information on how patients are actually doing.

Turning a Score Into an Action

From a therapeutic perspective, incorporating PROs can contribute to a more targeted conversation about issues of concern to the patient that might otherwise go unaddressed. Information on disease control — the number of asthma episodes, for instance — can focus a visit on the topics that matter most and serve as a useful measure of disease burden.

Having a prespecified, agreed-upon PRO score threshold that triggers a specific, evidence-based intervention facilitates action by the clinician. Alternatively, aggregate patient-reported data on symptoms, function, quality of life, or experience can be shared directly with the patient as part of a shared decision-making conversation.

PROS IN ACTION, 2 OF 6

Self-Management Support: The Model

Self-management support is a key component of the Chronic Care Model, a framework that has driven health system improvement. Informed, activated patients are key to the productive patient-clinician relationships that drive outcome improvement, through programs with four elements.

Clarifying Roles

Clinicians communicate the expectation that the patient plays a central role in managing their own condition.

Ongoing Assessment

Patients' self-management skills, confidence, barriers, and supports are assessed regularly.

Behavior Change Training

Trained staff employ behavior change interventions rather than simply giving instructions.

Co-Developed Plans

Patients co-develop individualized treatment plans with their care team, with support ongoing.

The Blind Spot Between Office Visits

Because medical care at outpatient visits happens at relatively infrequent intervals while health events occur continuously, the trajectory of a patient's illness may not be adequately captured at a clinic visit — due to failures of patient recall, a lack of any way to organize the data in between, and, on the clinician's side, simply not asking when PROs aren't being assessed.

Consider two patients starting new medications. Patient A begins Drug A and experiences rapid improvement in signs and symptoms, which then gradually return to baseline over time. Patient B begins Drug B and experiences gradual improvement, then a rapid worsening, before the condition returns to that same baseline by the follow-up visit. Both patients show roughly the same disease activity at that visit — so without between-visit data, the relative ineffectiveness of Drug B, and the additional time Patient B spent in a flare, are both invisible to the clinical decision.

Technology Fills the Gap

Technology can enable the capture and transmission of a patient's unique experience to the clinical team between visits. Data can be collected at home, either as PRO questionnaires on a computer or mobile device, or as technology-enabled patient-generated data such as physiologic monitors, then shared with the clinical team.

Numerous potential barriers exist: the availability of an optimal technology platform, the cost of that technology, and whether the patient has access to a reliable, persistent Wi-Fi connection. From the clinician's side, additional considerations include the technology's interoperability with the electronic health record, training staff to use the system and interpret results, and building review of the data into clinical workflow — because once patients are transmitting data, they expect it to be reviewed and acted on promptly.

PROS IN ACTION, 5 OF 6

Orchestra: Co-Producing Care With Mobile Health

Such a system is currently being tested in pediatric chronic illness care, with the Orchestra mobile health technology platform and care model intervention in cystic fibrosis and inflammatory bowel disease.

The concept, currently being piloted, is a mobile health platform that enables clinicians and patients to work collaboratively to co-produce healthcare — in part by allowing symptom tracking with real-time data visualization shared between patient and clinical team, continuing their interaction outside the office visit. The system contains automated symptom surveillance software that generates alerts when potentially important changes in health status are detected, and also supports previsit planning and opportunities for patients to participate in research.

What Success Looks Like

Operationalizing mobile health technology like Orchestra raises its own questions: the optimal way to integrate such a system into clinical workflow, and how to evaluate it for feasibility, acceptance, and completion rates. Embracing the co-production model requires a culture shift, with both internal and external stakeholder engagement and training.

Ultimately, success may be measured in terms of increased patient engagement and self-efficacy — a patient's belief in their own ability to perform the activities necessary to achieve their health goals — along with improved outcomes and appropriate healthcare utilization. The concept is potentially transformative for a patient's ability to self-manage care, including individualized treatment plans, customized treatment trials, and N-of-1 studies built around a patient's own PRO data.

RESEARCH CHALLENGES, 1 OF 7

Why PROs Matter for Pharmacoepidemiology

PROs are directly relevant to pharmacoepidemiology research because they capture information central to estimating the benefit and potential adverse effects of a medication that may elude a physical exam or lab test — symptoms like fatigue, cognitive impairment, or pain.

Clinical trials have a restricted sample size and duration of use compared to real-world consumption of medications. Routine, systematic collection of PROs as part of clinical care can help researchers understand the epidemiology of disease and, importantly, the impact of medications on symptoms and quality of life as part of postmarket surveillance — provided the data is reliable, drawn from a representative population, and accessible for research.

RESEARCH CHALLENGES, 2 OF 7

From Recognition to Action

While PRO collection benefits pharmacoepidemiology research, the underlying rationale for bringing the patient's voice into clinical care through PROs is to identify unmet needs and gaps in healthcare.

Once such a need is identified, ideally an action or response follows: a change in medication, a referral for treatment, or an individualized plan for lifestyle or behavior change as part of self-management support. Motivating that move from recognizing a problem to actually acting on it is a major area for attention — a PRO score that nobody acts on has not helped the patient at all.

RESEARCH CHALLENGES, 3 OF 7

Getting PRO Collection Right

To generate valid estimates of treatment effectiveness from PRO data, clinical processes must be in place to secure research-quality data — and the clinical team and patients must be aligned on why quality PRO collection matters, since the primary motivation is better patient-centered care, with research use as a secondary benefit.

To be most useful clinically, and to allow valid research inferences, PROs must be collected four ways: routinely, at reliable, regular intervals; uniformly, in a consistent manner; completely, so they can be scored; and correctly, so patients understand the questions and their answers actually reflect their health status.

RESEARCH CHALLENGES, 4 OF 7

When Clinicians Don't Review the Results

Once PROs are incorporated into clinical care, there must be a reliable means for the clinical team to review the results — and this doesn't always happen. One leading institution found a discordance between symptom documentation from self-report and a lack of matching clinician documentation for signs of heart disease.

Nonreview has real consequences: it threatens co-production of care, risks patients feeling their time and input weren't valued, and can decrease willingness to complete PROs at future visits. It is also a missed opportunity to check whether the collection system itself is working, and to verify that patients understand the instrument and that it reflects their health status.

RESEARCH CHALLENGES, 5 OF 7

Choosing the Right PRO, and Making Scores Actionable

Instrument selection is vitally important for both clinical and research use: a measure should be clinically relevant, address the outcome of interest, and be brief and practical to administer, with adequate reliability, validity, and responsiveness. Just as important, the measure should be chosen with representative patients and clinicians at the table — not on psychometric properties alone — weighing ease of understanding, usefulness for communication, and the ability to identify unmet needs. A comparison study of PRO measures in oncology, with feedback from both patients and clinicians, helped highlight exactly this point.

To be actionable, scores for a specific health condition need a known normal range and a known point at which to intervene for an abnormal score — which requires consensus among clinicians and patients on recommended thresholds and interventions. Knowing the minimal clinically important difference for improvement or worsening helps track longitudinal change, moving PRO use from simple review toward genuinely co-producing a treatment plan.

RESEARCH CHALLENGES, 6 OF 7

Individualized Care and N-of-1 Trials

Mobile health tracking systems, like the Orchestra example, offer the possibility of customized measurement — choosing which PROs to use and when to administer them — which can be of great importance in pharmacoepidemiology outcome studies.

Using PROs for continuous, in-between-visit monitoring, with customized reporting and a way to detect signals of changing health status in a personalized system, lends itself naturally to N-of-1 trials: an individual can make a planned change to treatment or lifestyle, then track symptoms to evaluate whether it actually helped. Keeping a daily journal or other record can also reveal whether other triggers or environmental factors explain a change in health status.

RESEARCH CHALLENGES, 7 OF 7

The Barriers to Making This Routine

Barriers to PRO use are varied, starting with the institutional resources needed to obtain, store, and update electronic data collection tools. Designing PRO integration into clinical workflow requires gaining clinician consensus on the purpose and value of adding PRO capture and review to the visit.

Clinicians need to understand how structured PROs add value over simply asking a patient how they feel, and frontline staff need training on collecting and distributing questionnaires reliably. Buy-in tends to become easier as more convincing evidence accumulates that PRO integration improves outcomes, as the logistics of collection get easier, as graphical displays become more intuitive, and as clinicians get feedback on their own patients' scores and see they can act on it.

METHODOLOGIC CHALLENGES, 1 OF 7

When Patients and Clinicians See Things Differently

Just as patient engagement and PROs inform discussion in the clinical setting, they play an increasing role in research too — including a priority of the FDA to include patient-relevant outcomes in clinical trials, and the genesis of PCORI's funding portfolio. But incorporating PROs into clinical research raises real methodologic challenges, starting with discordance in how patients, clinicians, and researchers perceive the same condition.

There are documented discrepancies between how clinicians and patients perceive a patient's level of disease activity, and whether a condition has improved or deteriorated. There can also be a lack of concordance between the composite measures used to assess efficacy in clinical trials and measures built from PROs.

METHODOLOGIC CHALLENGES, 2 OF 7

Case Study: Rheumatoid Arthritis and the DAS-28

In rheumatoid arthritis clinical trials, composite indices such as complete joint counts, laboratory values, and physician global assessment tend to be lengthy to complete compared to PROs, which are shorter and more feasible for everyday clinical practice.

Consider the DAS-28 (disease activity score), a common composite index that assesses 28 joints for tenderness or swelling — but does not include the feet or ankles. A patient with painful foot involvement may not be satisfied with the improvement the DAS-28 shows, because it excludes an important part of their disease experience. Using the DAS-28 as an outcome in pharmacoepidemiology studies could therefore miss what matters to patients and produce biased measures of association.

METHODOLOGIC CHALLENGES, 3 OF 7

Composite Scores vs. the Patient's Experience

Studies have found that while composite indices and PRO results are correlated, they do not always track together. When a PRO is an independent predictor of treatment response, its results should be reported separately rather than folded into a composite index — supporting the case for using PROs alongside established composite measures rather than instead of them.

The same investigators studying rheumatoid arthritis found that patients required greater improvement in their condition than physicians did before reporting satisfaction, and needed less deterioration than physicians did before reporting dissatisfaction. Concordance between patients and clinicians on whether disease activity had changed was only about 60 percent — a reminder that the patient's own viewpoint may not always align with what a research study is measuring.

METHODOLOGIC CHALLENGES, 4 OF 7

Measuring Whether a Change Really Matters

Understanding how to estimate clinically meaningful change using PROs is important for determining whether a treatment actually works — and it is complex. Standard statistical approaches can detect a statistically significant difference in scores over time, but that detection does not by itself mean the change is clinically important.

What counts as a meaningful difference can depend on who is asked — patient, caregiver, or clinician — on the health condition being studied, and on whether the patient is improving or getting worse at the time it is measured. Instability in a rater's own judgment, changes tied to disease stage, and bias in recalling one's own past state all make this classification genuinely difficult.

METHODOLOGIC CHALLENGES, 5 OF 7

Two Ways to Set a Meaningful Threshold

Two related terms describe the smallest PRO score change worth caring about. The minimal important difference (MID) is the smallest change that could be determined important; the minimal clinically important difference (MCID) is generally the smallest difference that would prompt an actual change in a patient's management, determined using clinical anchors. In practice the two are sometimes used interchangeably, though their estimates can differ.

Anchor-Based Methods

Use an external indicator considered clinically relevant — a lab test, a clinician rating, or a patient's own global rating of change — to place a patient's change on a continuum from large negative to large positive. Ideally, multiple relevant anchors are used across multiple samples to confirm the estimate; receiver operating characteristic (ROC) curves are another anchor-based technique for identifying meaningful improvement or worsening.

Distribution-Based Methods

Based purely on statistics, without a clinical anchor: express the effect in standard deviation units or the standard error of measurement, useful when anchor methods aren't available.

METHODOLOGIC CHALLENGES, 6 OF 7

Bookmarking: Learning Thresholds From Stakeholders

An alternative approach, borrowed from educational testing, presents a representative panel of stakeholders — patients, caregivers, clinicians — with clinical vignettes spanning a continuum of scores, and asks the panel to place a “bookmark” separating different levels of severity.

For physical function, for example, vignettes might range from no limitation through severe disability; the panel orders them and sets thresholds between “no problem,” “mild problem,” “moderate problem,” and “severe problem.” A related scale judgment method has raters compare a patient's questionnaire answers from before and after treatment and judge whether an important difference occurred. Both approaches require real qualitative work across different patient populations to build agreement on where the thresholds should sit.

METHODOLOGIC CHALLENGES, 7 OF 7

When the Goalposts Move: Response Shift

A further complication is that a person's own perception or valuation of what's being measured can change over time — a phenomenon called “response shift.” It can occur through a change in a patient's internal standard of measurement, a change in what they value, or a redefinition of the thing being measured altogether.

This might happen if a psychological intervention changed how a patient thinks about their condition, or if a patient simply adapts to disease progression or gets used to medication side effects over time. Response shift can alter estimates of treatment effect over time, and may be at the root of cases where a patient's own global rating of change doesn't match what other assessments would predict.

Practical Guides Are Emerging

With the growing interest in and adoption of PROs, publications now offer guidance on training clinicians in PRO use and on presenting PRO scores to patients and clinicians. Recommendations include eliciting clinicians' own concerns during training, keeping training brief and experiential with case studies, and including relevant decision aids.

Useful public resources include the International Society for Quality of Life Research's user's guide to implementing PRO assessment in clinical practice, the National Quality Forum's guide to PROs in performance measurement, a PCORI-funded guide to integrating PROs into electronic health records, and the Health Measures website, a repository for PROMIS and other NIH-funded measures such as Neuro-QoL, the NIH Toolbox, and ASCQ-Me.

Presenting PRO Data Well

How PRO scores are displayed graphically turns out to matter. Preferences vary by audience: some studies find a preference for line graphs, others for bar graphs, and patients tend to prefer simpler formats while clinicians tend to prefer more data.

Directionality also matters: research shows that portraying better health as higher on the chart, and including reference lines for normal versus abnormal thresholds, helps patients and clinicians interpret scores correctly. As the field matures, more electronic health record vendors are expected to build in graphical displays like these by default.

A Cultural Shift Already Underway

Overcoming the perceived barriers to PRO adoption in clinical care may take a genuine cultural shift — including accepting co-production as a way to transform healthcare delivery and improve chronic illness care.

For some clinicians, it helps to view PROs as a tool for detecting a meaningful signal in a patient's health, or as a more efficient and accurate way to exchange information with a patient. As evidence mounts that PRO-supported co-production improves quality of care and outcomes, the tide may continue shifting toward embracing PROs as a standard part of clinical care.

The Future: PROs Meet Big Data

PROs are likely to become increasingly used in comparative effectiveness research, with patient engagement in study design leading to more meaningful use of PRO measures, and PRO capture increasingly built into learning health networks and disease registries.

There is growing interest in using PRO data captured in electronic health records as a structured outcomes source for comparative effectiveness research — potentially powerful when combined with other electronic data sources, and especially useful when no standardized outcome assessment exists in the clinical note. Such data could ultimately let clinical registries be combined with administrative claims data to support research at much larger scale.

The Future: Wearables and Physiology

An emerging area of development is the use of wearable devices with health monitors. Previously used mainly as sports fitness trackers, wearables can now track heart rhythms with electrocardiograms transmitted to medical facilities, and blood glucose monitoring may eventually become an option too.

Coupling physiologic data with PRO data will be informative for understanding what a change in score actually means — for example, helping a patient better calibrate a worsening feeling of fatigue, as captured in a PRO score, to a corresponding physiologic change, keeping individual patients more in tune with how their symptoms and function reflect what's happening in their body. An important unresolved question, the chapter notes, is how best to analyze and interpret longitudinal PRO data, and which models of continuous PRO collection will prove scalable and generalizable across different settings.

Key Takeaways

Patient engagement reframes healthcare as something clinician and patient co-produce together, and patient-reported outcomes are the tool that makes the patient's own voice systematic, structured, and usable — in a clinic visit, a clinical trial, and an observational study alike.

PROs pay off across the care continuum: they sharpen what a new drug is measured against, they help clinicians catch what a chart alone would miss, and — as the oncology survival finding shows — reviewing and acting on them can improve outcomes that matter, not just processes.

Making PROs work is still genuinely hard: they must be collected reliably and actually reviewed, composite research measures can miss what matters most to an individual patient, and researchers still lack full consensus on how to measure whether a change in score is a change worth caring about.

WRAPUP, 2 OF 2

Thank You

Patient engagement and patient-reported outcomes sit at the intersection of clinical care and clinical research — a reminder that the ultimate measure of whether a treatment works is whether it makes a meaningful difference to the person receiving it.

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